

● HARD

● INFORMATION AND IDEAS

● ANSWER KEY

Inferences

07

1 answers · Rationale-rich · Teach from doc

Q1**D**

- D. aerospace composites containing the nitrogen-treated SiC fiber may have the ability to withstand mechanical stress for a longer period of time than can aerospace composites containing either of the two polymer-derived SiC fibers.

Choice D is the best answer because it most logically completes the text's discussion of silicon carbide (SiC) fibers and creep, or deformation related to ongoing mechanical stress and elevated temperatures. The text states that Bhatt et al. found that a nitrogen-treated SiC fiber had a lower minimum creep rate than two polymer-derived SiC fibers did. Because having a lower creep rate means that the material is slower to deform with exposure to stress, as the text explains, this finding suggests that aerospace composites made with the nitrogen-treated SiC fiber may be able to withstand mechanical stress for a longer period than those made with the other two polymer-derived SiC fibers can.

Choice A is incorrect because it overstates the implications of the study's findings, which have to do with the rate of a material's deformation under stress, not the absolute degree of deformation. The text states that Bhatt et al. observed that a nitrogen-treated SiC fiber had a lower minimum creep rate than two polymer-derived SiC fibers did, meaning only that it deformed more slowly over time under constant stress, not that it underwent less deformation overall. Choice B is incorrect because the text doesn't establish any similarity between the two polymer-derived SiC fibers other than that both had a higher creep rate than the nitrogen-treated SiC fiber did in Bhatt et al.'s study. Moreover, reducing a material's resistance to creep would mean that the material becomes *more* susceptible to deformation with exposure to stress and elevated temperatures, which would be expected to shorten rather than prolong the lifespan of machinery made with that material. Choice C is incorrect because the text suggests that the stability of aerospace equipment may be better improved by composites containing nitrogen-treated SiC fiber than by composites containing the two polymer-derived SiC fibers, not the other way around. The text indicates that Bhatt et al. observed that the nitrogen-treated SiC fiber had a lower minimum creep rate than the other two fibers did, meaning that it was slower to degrade under exposure to mechanical stress and elevated temperatures—suggesting that it may remain stable for longer periods.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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